TILE = 8

```
GPU: NVIDIA GeForce RTX 4050 Laptop GPU (compute capability 8.9)

     N |    CPU(ms) |  Naive(ms) |  Tiled(ms) |   Naive x |   Tiled x |  Tiled GF |   MaxDiff
-------+------------+------------+------------+-----------+-----------+-----------+----------
   256 |       8.89 |      0.094 |      0.076 |     94.1x |    116.5x |     439.8 |  2.29e-05
   512 |      59.05 |      0.674 |      0.556 |     87.7x |    106.2x |     482.6 |  4.58e-05
  1024 |     472.22 |      5.267 |      4.342 |     89.6x |    108.8x |     494.6 |  9.16e-05
  2048 |    4831.09 |     42.031 |     34.177 |    114.9x |    141.4x |     502.7 |  1.83e-04

Done.

C:\Users\liang\source\repos\GPU Matrix Multiplication\x64\Release\GPU Matrix Multiplication.exe (process 23120) exited w
ith code 0 (0x0).
Press any key to close this window . . .
```

TILE = 16

```
GPU: NVIDIA GeForce RTX 4050 Laptop GPU (compute capability 8.9)

     N |    CPU(ms) |  Naive(ms) |  Tiled(ms) |   Naive x |   Tiled x |  Tiled GF |   MaxDiff
-------+------------+------------+------------+-----------+-----------+-----------+----------
   256 |       7.90 |      0.074 |      0.060 |    106.7x |    131.4x |     558.2 |  2.29e-05
   512 |      62.87 |      0.553 |      0.425 |    113.6x |    148.1x |     632.3 |  4.58e-05
  1024 |     488.26 |      4.247 |      3.256 |    115.0x |    149.9x |     659.5 |  9.16e-05
  2048 |    5082.69 |     34.129 |     24.490 |    148.9x |    207.5x |     701.5 |  1.83e-04

Done.

C:\Users\liang\source\repos\GPU Matrix Multiplication\x64\Release\GPU Matrix Multiplication.exe (process 14860) exited w
ith code 0 (0x0).
Press any key to close this window . . .
```

TILE = 32\

```
GPU: NVIDIA GeForce RTX 4050 Laptop GPU (compute capability 8.9)

     N |    CPU(ms) |  Naive(ms) |  Tiled(ms) |   Naive x |   Tiled x |  Tiled GF |   MaxDiff
-------+------------+------------+------------+-----------+-----------+-----------+----------
   256 |       9.50 |      0.090 |      0.071 |    105.5x |    134.6x |     475.6 |  2.29e-05
   512 |      56.54 |      0.549 |      0.438 |    103.0x |    129.0x |     612.5 |  4.58e-05
  1024 |     493.28 |      4.303 |      3.345 |    114.6x |    147.5x |     642.0 |  9.16e-05
  2048 |    4990.91 |     33.212 |     22.943 |    150.3x |    217.5x |     748.8 |  1.83e-04

Done.

C:\Users\liang\source\repos\GPU Matrix Multiplication\x64\Release\GPU Matrix Multiplication.exe (process 14200) exited w
ith code 0 (0x0).
Press any key to close this window . . .
```

Results and analysis. I benchmarked a single-threaded CPU implementation against a naive CUDA kernel and a shared-memory tiled kernel on an RTX 4050 Laptop GPU (compute capability 8.9), at matrix sizes from 256 to 2048 and tile sizes of 8, 16, and 32. At N = 2048, the best configuration (TILE = 32) ran in 22.9 ms, a 218x speedup over the CPU baseline and 1.45x faster than the naive GPU kernel, reaching about 749 GFLOPS. All configurations matched the CPU output to within a max absolute difference of 1.83e-4. Performance improved with tile size but with diminishing returns: moving from TILE = 8 to 16 cut the tiled kernel's time by 28%, while moving from 16 to 32 cut it by only 6%. Larger tiles raise data reuse (each value loaded into shared memory is used TILE times, cutting global memory loads to 1/TILE of the naive kernel) and give better warp utilization and memory coalescing, but gains flatten as the kernel moves from memory-bound toward compute-bound. The naive kernel also sped up with larger block sizes (42.0 ms to 33.2 ms), since block shape affects how well a warp's memory accesses coalesce even without shared memory. Tiling's measured advantage over naive is modest (1.2x to 1.45x) compared with the theoretical reduction in global loads, likely because the GPU's L1/L2 caches already absorb much of the naive kernel's repeated reads. My kernel reaches well below the GPU's theoretical peak, and a vendor-tuned library like cuBLAS would likely be much faster, so further gains would come from register blocking and vectorized loads. Timings use CUDA events averaged over 10 runs after a warm-up, with a single CPU run per size.